Metodologias de Investigação
Research Methodology

Nuno Feixa Rodrigues

nfr@ipca.pt

Instituto Politécnico do Cávado e do Ave

Escola Superior de Tecnologia

Office hours

- Gabinete 3 EST
- Terça-feira
  - 16:00 – 18:00
- Sexta-feira
  - 18:00 – 20:00 (Online)

Objectives

- **To learn, think, write and talk (scientifically)!**
- Assimilation of methodologies, processes and practices related to scientific research
- Development of critical spirit through scientific thinking skills.
- Development of the ability to produce a scientific text.
- Define a research design and work plan.
- Communication of scientific content.

Syllabus

1. Theoretical component:
   1. Understand the concepts of scientific research
   2. Identify a problem and carry out the research literature review
   3. To define a conceptual model and hypotheses
   4. To define research methodology and work plan
   5. Scientific writing
2. Practical component:
   1. Research and analysis of scientific papers
   2. Written of a literature review scientific paper
   3. Develop of the work plan of the master dissertation

Bibliography

- **Principal**
  - J. David Creswell, John W. Creswell (2018), Research Design: Qualitative, Quantitative, and Mixed Methods Approaches (5th Ed.), SAGE Publications.
  - Dey, R., Deb, D., Balas, V. E. (2019), Engineering Research Methodology: A Practical Insight for Researchers, Springer Singapore.
  - Hair, Joseph F, Jr., Rolph E. Anderson, Ronald L. Tatham e William C. Black (1998), Multivariate Data Analysis (5th Ed.), Upper Saddle River, US: Prentice Hall.
  - Paul R. Cohen, P.R. (1995), Empirical Methods for Artificial Intelligence (Bradford Books) (1st Ed.), The MIT Press, USA.
  - Yin, R.K. (1989), Case Study Research. Design and Methods, Newbury Park: SAGE Publication, California.
- **Complementary**
  - Ladyman, J. (2012). ***Understanding philosophy of science***. Routledge.
  - Hair, Joseph F, Jr., Rolph E. Anderson, Ronald L. Tatham e William C. Black (1998) Multivariate Data Analysis (5th Ed.); Upper Saddle River, US: Prentice Hall.
  - Patrick F. Dunn (2010), Measurement and Data Analysis for Engineering and Science, CRC Press, New York.

Course work and evaluation

- During the course students will work to accomplish the following outcomes.
  - D1: A review paper about a subject, problem or hypothesis of interest.
  - D2: A research plan for the MSc work.
- Deadlines
  - **D1: 18/04/2026**
  - **D2: 16/05/2026**
- Public presentation of D1 and D2 will take place at the first class after each deadline.

Milestones

- D1 Literature Review
  - **28/02** – Research question & search query
  - **07/03** – Selection and application of selection criteria
  - **14/04** – Data extraction
  - **18/04** – Research article completed
- D2 Research Plan
  - **25/04** – Research question
  - **02/05** – Methods
  - **16/05** – Final research plan
- Public presentations of each deliverable take place on the first class after each delivery date.

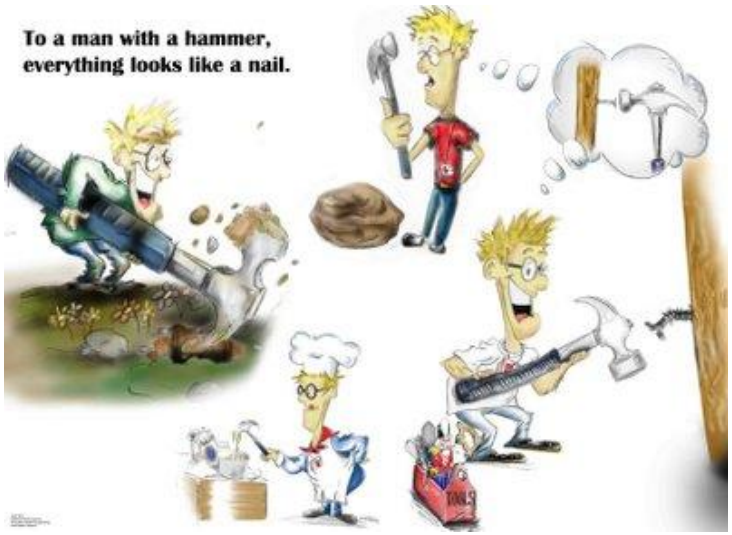
Course work and evaluation

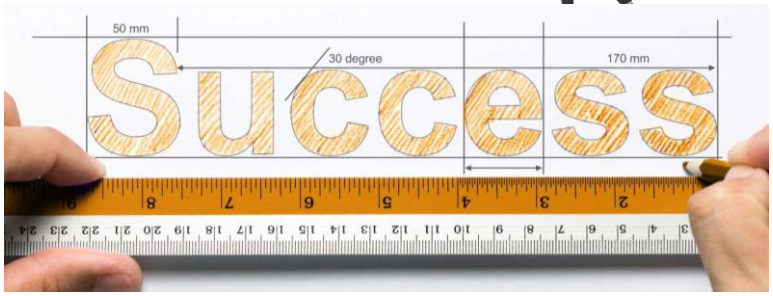
- The final grade is calculated according to the following expression.
  - $FG = D1 \times 0.7 + D2 \times 0.3$
  - Where $D1 >= 10$ and $D2 >= 10$
- D1 and D2 will be evaluated according to the following criteria
  - Completeness
  - Quantitative and qualitative analysis
  - Research methodology and design
  - Text quality
  - Oral presentation
- It is not allowed to perform (or improve) the curricular unit in the exam season, special season or exceptional exam season.

Master of Science of What?

Problem

- We start with **problems**, not with solutions.
  - *"I want to use deep learning to guide robots in a factory."*
- Problems need to be specific
  - *"I want to improve maintenance in manufacturing."*
- Problems need to be observable and measurable.
  - *"I want to develop a industry 4.0 smart system to integrate mining machines."*





Problem

- *"Use deep learning to guide robots in a factory"*
  - **"Current robot navigation in Factory X fails when corridors layout changes, causing X% downtime."**
- *"Improve maintenance in manufacturing"*
  - **"Reduce unplanned downtime caused by bearing failures in CNC machines in small-scale manufacturing plants."**
- *"I want to develop a industry 4.0 smart system to integrate mining machines."*
  - **"Operators of mining tractors lack real-time indicators of the position of other machines near them to prevent collisions."**

Research Questions

- Current robot navigation in Factory X fails when corridors layout changes, causing X% downtime.
  - RQ1: What **types of corridor layout changes** most frequently lead to robot navigation failures in Factory X?
  - RQ2: How does **navigation performance** degrade as a function of the **frequency and magnitude** of **corridor** layout **changes**?
  - RQ3: Which **environmental** or **system factors** most strongly influence the robot's ability to **recover** from unexpected layout changes?

Research Questions

- Reduce unplanned downtime caused by bearing failures in CNC machines in small-scale manufacturing plants.
  - RQ1: Which measurable operational **signals precede bearing failures** in CNC machines in small-scale manufacturing plants?
  - RQ2: How **early can bearing failures be reliably detected** using available machine and sensor data?
  - RQ3: What is the **impact of earlier bearing failure** detection on unplanned downtime in small-scale manufacturing environments?

Research Questions

- Operators of mining tractors lack real-time indicators of the position of other machines near them to prevent collisions.
  - RQ1: **What spatial awareness information** do mining tractor operators currently lack during machine operation?
  - RQ2: How **accurately** can **the relative position of nearby machines** be estimated under typical mining site conditions?
  - RQ3: How does access to **real-time proximity information** affect operator response time in potential **collision scenarios**?

Good research questions

- Avoid naming a solution or method.
- Can be answered with data, experiments, or observations.
- Naturally lead to hypotheses and evaluation metrics.

Hypothesis

- Current robot navigation in Factory X fails when corridors layout changes, causing X% downtime.
  - H: Navigation failures increase significantly after corridor layout changes because the **robot's internal environment representation does not update fast enough** to reflect new spatial constraints.

- Reduce unplanned downtime caused by bearing failures in CNC machines in small-scale manufacturing plants.
  - H: Bearing failures in CNC machines are preceded by detectable changes in **vibration and temperature patterns** that occur early enough to allow intervention before unplanned downtime.

- Operators of mining tractors lack real-time indicators of the position of other machines near them to prevent collisions.
  - H: Providing operators with real-time proximity **information about nearby machines reduces reaction time** in potential collision scenarios compared to operation without such information.

Good Hypothesis

- Directly **answers one aspect** of the stated problem.
- Is **falsifiable.**
- Makes a **bounded causal** claim.
- Avoids naming a specific **technology** or **method.**
- Can be evaluated with **realistic** (MSc-scale) experiments or data.

From Hypothesis to Methods

- **Problem:** Operators of mining tractors lack real-time indicators of the position of other machines near them to prevent collisions.
- **Hypothesis**: Providing real-time proximity information reduces operator reaction time.
- **Variables**
  - **Independent**: availability of proximity indicators
  - **Dependent**: reaction time, number of near-collisions
- **Methods**
  - Simulator-based experiment with scripted collision scenarios
  - Controlled field study with and without indicators
  - Logged operational data comparison (before/after deployment)

From Methods to Evaluation…

- **Evaluation**
  - Statistical comparison of reaction times
  - Analysis of incident frequency
  - Operator performance under identical conditions
- **Decision**
  - Hypothesis **supported** or **rejected**